The Chinese University of Hong Kong, Shenzhen
School of Data Science

MAT3007 · Homework 3

Due: 11:59.59 pm, Oct 20, 2021

Instructions:

- Homework problems must be carefully and clearly answered to receive full credit. Complete sentences that establish a clear logical progression are highly recommended.
- You must submit your assignment in Blackboard. If some problems need coding, you should paste your code in the file. Please upload only one file(pdf). The file name should be in the format **last name-first name-hw2. Any nonstandard assignment will not be graded**..
- The homework must be written in English.
- **Late submission will not be graded.**
- Each student **must not copy** homework solutions from another student or from any other source.

Problem 1 (5+5+10=20pts).
Consider the following linear program:

$$\begin{array}{lll} \text{maximize} & x_1 + 3x_2 + 3x_3 + 8x_4 & \\ \text{subject to} & x_1 - x_2 + x_3 & \leq 2 \\ & x_3 - x_4 & \leq 1 \\ & 2x_2 + 3x_3 + 4x_4 & \leq 10 \\ & x_1, x_2, x_3, x_4 & \geq 0. \end{array}$$

(a) Transform this problem to standard form and find a trivial BFS.

(b) Compute the reduced costs $\bar{\mathbf{c}}$.

(c) Choose the incoming basis according to **Bland's Rule**, compute the step size θ^* and and jth basic direction, then apply the simplex method to update the current BFS to a neighboring one.

Solution.

(a) We first express the problem in standard form:

$$\begin{array}{lll} \text{minimize} & -x_1 - 3x_2 - 3x_3 - 8x_4 & \\ \text{subject to} & x_1 - x_2 + x_3 + s_1 & = 2 \\ & x_3 - x_4 + s_2 & = 1 \\ & 2x_2 + 3x_3 + 4x_4 + s_3 & = 10 \\ & x_1, x_2, x_3, x_4, s_1, s_2, s_3 & \geq 0. \end{array}$$

Since the right-hand side is nonnegative, we can choose the slack variables as basic variables and set the original variables to zero, i.e., the point $(x; s) = (0, 0, 0, 0, 2, 1, 10)^\top$ is a valid initial BFS with basis $B = \{5, 6, 7\}$.

(b) The reduced costs of different non-basic variables can be computed by $\bar{c}_j = c_j - c_b^T A_B^{-1} A_j$. At the current point, c_b is a zero vector, so we easily get $\bar{c} = [\bar{c}_1, \bar{c}_2, \bar{c}_3, \bar{c}_4]^T = [c_1, c_2, c_3, c_4]^T = [-1, -3, -3, -8]^T$.

(c) According to **Bland's Rule**, we choose x_1 as the incoming variable. Then we can compute the basic direction by $d_j = [d_B; d_N] = [-A_B^{-1} A_j; 0; ...; 1; ...; 0]$. At the current point, A_B is an identity matirx, so the calculations are easy. $d_B = -I^{-1} A_1 = -A_1 = [-1, 0, 0]^T, d_N = [1, 0, 0, 0]^T$. Then, the step size can be derived by $\theta^* = min_{i \in B, d_i < 0}\{-x_i/d_i\}$. The minimun is obtained at $-x_5/d_5 = 2$ (we use x_5 here to represent s_1), which means the outgoing variable is s_1. Then, the new BFS is $x + \theta^* d = [0, 0, 0, 0, 2, 1, 10]^T + 2 * [1, 0, 0, 0, -1, 0, 0]^T = [2, 0, 0, 0, 0, 1, 10]^T$ with basis $B = \{1, 6, 7\}$. And the objective value is reduced by $\theta^* \bar{c}_1 = 2 * -1 = -2$.

Problem 2 (20pts). Consider the same linear program in **Problem 1**, please use simplex tableau to completely solve it. For each step, draw the simplex tableau. Clearly mark what is the current basis, the current basic solution, and the corresponding objective function value. You can start from the same trivial BFS found in **Problem 1**. Also, compare the BFS obtained in **Problem 1** and the second tableau obtained in this question. They should be consistent.

Solution.
We first express the problem in standard form:

$$\begin{aligned}
\text{minimize} \quad & -x_1 - 3x_2 - 3x_3 - 8x_4 \\
\text{subject to} \quad & x_1 - x_2 + x_3 + s_1 = 2 \\
& x_3 - x_4 + s_2 = 1 \\
& 2x_2 + 3x_3 + 4x_4 + s_3 = 10 \\
& x_1, x_2, x_3, x_4, s_1, s_2, s_3 \geq 0.
\end{aligned}$$

Since the right-hand side is nonnegative, we can choose the slack variables as basic variables and set the original variables to zero, i.e., the point $(x; s) = (0, 0, 0, 0, 2, 1, 10)^\top$ is a valid initial BFS with basis $B = \{5, 6, 7\}$. Notice that the initial tableau is already in canonical form, i.e., we have:

B	−1	−3	−3	−8	0	0	0	0
5	1	−1	1	0	1	0	0	2
6	0	0	1	−1	0	1	0	1
7	0	2	3	4	0	0	1	10

and the current objective function value is 0. The pivot column is {1}; the pivot row is {5}; the pivot element is 1; after the row updates we obtain the new tableau:

B	0	−4	−2	−8	1	0	0	2
1	1	−1	1	0	1	0	0	2
6	0	0	1	−1	0	1	0	1
7	0	2	3	4	0	0	1	10

The basis updated to $\{1,6,7\}$; the current BFS is $(x;s)=(2,0,0,0,0,1,10)^\top$; the current objective function value is -2. The pivot column in the updated tableau is $\{2\}$; the pivot row is $\{7\}$; the pivot element is 2; after the row updates the new tableau is given by:

B	0	0	$\frac{5}{2}$	0	1	0	2	22
1	1	0	$\frac{5}{2}$	2	1	0	$\frac{1}{2}$	7
6	0	0	1	-1	0	1	0	1
2	0	1	$\frac{3}{2}$	2	0	0	$\frac{1}{2}$	5

Since all reduced costs are nonnegative, the method stops with the optimal solutions $x^*=(7,5,0,0)^\top$ and $s^*=(0,1,0)^\top$; the optimal basis is $B=\{1,2,6\}$ and the optimal value (of the original problem) is 22.

Problem 3 (20pts).
Use the two-phase simplex method (implemented by simplex tableau) to completely solve the linear optimization problem. For each step, draw the simplex tableau. Clearly mark what is the current basis, the current basic solution, and the corresponding (negative) objective function value.

$$\begin{aligned} \text{minimize} \quad & x_1+3x_2+x_4-2x_5 \\ \text{subject to} \quad & x_1+2x_2+x_4+x_5 = 2 \\ & x_1+2x_2-6x_4+x_5 = 2 \\ & x_1+4x_2-3x_3+x_4 = -1 \\ & x_1,x_2,x_3,x_4,x_5 \geq 0. \end{aligned}$$

Solution.
The problem is already in standard form; we can immediately start with phase I.
Phase I. We first generate the auxiliary problem

$$\begin{aligned} \text{minimize} \quad & y_1+y_2+y_3 \\ \text{subject to} \quad & x_1+2x_2+x_4+x_5+y_1 = 2 \\ & x_1+2x_2-6x_4+x_5+y_2 = 2 \\ & -x_1-4x_2+3x_3-x_4+y_3 = 1 \\ & x_1,x_2,x_3,x_4,x_5,y_1,y_2,y_3 \geq 0, \end{aligned}$$

An initial BFS of the auxiliary problem is given by $(x;y)=(0,0,0,0,0,2,2,1)^\top$ with $B=\{6,7,8\}$. The reduced costs for the non-basic variables can again be calculated by $c_N-c_B^\top A_B^{-1}A_N=c_N-c_B^\top A_N$ since $A_B=I_m$. We have

$$\big(0,0,0,0,0\big)-\big(1,1,1\big)\begin{pmatrix} 1 & 2 & 0 & 1 & 1 \\ 1 & 2 & 0 & -6 & 1 \\ -1 & -4 & 3 & -1 & 0 \end{pmatrix}=\big(-1,0,-3,6,-2\big)$$

and the simplex tableau is given by:

B	-1	0	-3	6	-2	0	0	0	-5
6	1	2	0	1	1	1	0	0	2
7	1	2	0	-6	1	0	1	0	2
8	-1	-4	3	-1	0	0	0	1	1

The pivot column is $\{1\}$; the outgoing column is $\{6\}$; the pivot element is 1; after the row updates we obtain the new tableau:

B	0	2	−3	7	−1	1	0	0	−3
1	1	2	0	1	1	1	0	0	2
7	0	0	0	−7	0	−1	1	0	0
8	0	−2	3	0	1	1	0	1	3

The pivot column is $\{3\}$; the outgoing column is $\{8\}$; the pivot element is 3; after the row updates we obtain the new tableau:

B	0	0	0	7	0	2	0	1	0
1	1	2	0	1	1	1	0	0	2
7	0	0	0	−7	0	−1	1	0	0
3	0	$-\frac{2}{3}$	1	0	$\frac{1}{3}$	$\frac{1}{3}$	0	$\frac{1}{3}$	1

Since the reduced costs are nonnegative and the costs are zero, phase I of the two-phase simplex method stops with the BFS $(x; y) = (2, 0, 1, 0, 0, 0, 0)^\top$ and the basis $B = \{1, 3, 7\}$. This basis still contains the auxiliary variable y_2. We substitute it with x_4. Notice that the fourth column is the only possible choice since column $\{2\}$ and $\{5\}$ do not form a linearly independent set of vectors with columns $\{1\}$ and $\{3\}$. In order to maintain the canonical form, we perform one additional row update to change the fourth column to a unit vector. We obtain:

$$A_B^{-1}A = \begin{pmatrix} 1 & 2 & 0 & 0 & 1 \\ 0 & -\frac{2}{3} & 1 & 0 & \frac{1}{3} \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}, \quad c^\top - c_B^\top A_B^{-1}A = \begin{pmatrix} 0, 1, 0, 0, -3 \end{pmatrix}.$$

Phase II. Setting $B = \{1, 3, 4\}$, the initial costs are $-c_B^\top x_B = -2$. The corresponding initial simplex tableau is then given by:

B	0	1	0	0	−3	−2
1	1	2	0	0	1	2
4	0	0	0	1	0	0
3	0	$-\frac{2}{3}$	1	0	$\frac{1}{3}$	1

The pivot column is $\{5\}$; the outgoing column is $\{1\}$; the pivot element is 1; after the row updates we obtain the new tableau:

B	3	7	0	0	0	4
5	1	2	0	0	1	2
4	0	0	0	1	0	0
3	$-\frac{1}{3}$	$-\frac{4}{3}$	1	0	0	$\frac{1}{3}$

The method stops with the optimal solution $x = (0, 0, \frac{1}{3}, 0, 2)^\top$ and the optimal value -4.

Problem 4 Conditions for a Unique Optimum (10+10=20pts).
Let x^* be a basic feasible solution associated with some basic indices B. Prove the following:

a) If the reduced cost of every non-basic variable is positive, then x^* is the unique optimal solution.

Hint: Let $y \in \{x \in \mathbb{R}^n : Ax = b,\, x \geq 0\}$ be given with $y \neq x^*$. First, show that there exists $\ell \in N = B^{\mathsf{C}}$ such that $y_\ell > 0$. You can then mimic the proof of the theorem of stopping criterion based on the reduced costs.

b) If x^* is the unique optimal solution and if x^* is nondegenerate, then the reduced cost of every non-basic variable is positive.

Hint: The construction of the simplex method via basic directions can be helpful.

Solution.

(a) We can mimic the proof of the theorem in lecture: Let x be a BFS with $\bar{c}_N > 0$ and $\bar{c}_B = 0$ where $N = B^{\mathsf{C}}$ denotes the set of non-basic indices. Let $y \in P := \{x : Ax = b,\, x \geq 0\}$ be arbitrary with $y \neq x$ and define $d = y - x$. Then, as shown in the lecture, we have $Ad = Ay - Ax = 0$ and

$$0 = \begin{bmatrix} A_B & A_N \end{bmatrix} \begin{pmatrix} d_B \\ d_N \end{pmatrix} = A_B d_B + A_N d_N = A_B d_B + \sum_{j \in N} A_j d_j.$$

This implies $d_B = -\sum_{j \in N} A_B^{-1} A_j d_j$ and

$$c^\top d = \begin{pmatrix} c_B^\top & c_N^\top \end{pmatrix} \begin{pmatrix} d_B \\ d_N \end{pmatrix} = c_B^\top d_B + \sum_{j \in N} c_j d_j = \sum_{j \in N} (c_j - c_B^\top A_B^{-1} A_j) \cdot d_j = \sum_{j \in N} \bar{c}_j d_j.$$

We now have $\bar{c}_j > 0$ and $d_N = y_N - x_N = y_N \geq 0$ for all $j \in N$. If there exists $\ell \in N$ with $y_\ell > 0$, then our calculations show

$$c^\top d = c^\top y - c^\top x = \sum_{j \in N} \bar{c}_j d_j \geq \bar{c}_\ell y_\ell > 0.$$

Next, let us suppose $y_N = 0$ (then such an index ℓ does not exist). However, in this case the equality constraints yield

$$b = \begin{bmatrix} A_B & A_N \end{bmatrix} \begin{pmatrix} y_B \\ 0 \end{pmatrix} = A_B y_B \quad \Longrightarrow \quad y_B = A_B^{-1} b.$$

But then $x = y$ which contradicts our assumption $y \neq x$. Hence, for all $y \in P$ with $y \neq x$ we can infer that there exists $\ell \in N$ with $y_\ell > 0$. Thus, together, we obtain:

$$c^\top y - c^\top x > 0 \quad \forall\, y \in P \backslash \{x\}$$

and consequently, x is a *strict global solution* of the problem. Since strict global minima are unique, this finishes the proof.

(b) We follow the construction of the simplex method. In particular, let d be the j-th basic direction

$$d = \left[-A_B^{-1}A_j; 0; ...; 1; 0; ...; 0\right]^\top$$

As shown in the lecture, we then have $A(x + \theta d) = b$ for all $\theta \in \mathbb{R}$ and $x_i + \theta d_i = \theta d_i \geq 0$ for all $i \in N$ and $\theta \geq 0$. Since x is nondegenerate, we know that $x_i > 0$ for all $i \in B$. Consequently, there exists $\bar{\theta} > 0$ sufficiently small such that $x_i + \bar{\theta} d_i > 0$ for all $i \in B$. This implies $x + \bar{\theta} d$ is feasible and $x + \bar{\theta} d \neq x$. Since x is the unique global solution, it also needs to be a strict global solution (otherwise, if there exists another feasible point that achieves the same objective function value, this point would also be a global solution). As before, we obtain:

$$0 < c^\top (x + \bar{\theta} d) - c^\top x = \bar{\theta}[c_B^\top d_B + c_N^\top d_N] = \bar{\theta}[c_j - c_B^\top A_B^{-1} A_j] = \bar{\theta}\bar{c}_j.$$

Due to $\bar{\theta} > 0$, this yields $\bar{c}_j > 0$ and since j was arbitrary, we can infer $\bar{c}_N > 0$.

Problem 5 dual problem (10+10=20pts). Consider the following linear program:

$$\begin{array}{rc} \text{maximize} & 3x_1 + x_2 + 4x_3 \\ \text{subject to} & x_1 + 3x_2 + x_3 \leq 5 \\ & x_1 + 2x_2 + 2x_3 \leq 8 \\ & x_1, x_2, x_3 \geq 0 \end{array}$$

(a). What is the corresponding dual problem?

(b). Solve the dual problem graphically.

Solution.

(a). The dual problem is

$$\begin{array}{rc} \text{minimize} & 5y_1 + 8y_2 \\ \text{s.t.} & y_1 + y_2 \geq 3 \\ & 3y_1 + 2y_2 \geq 1 \\ & y_1 + 2y_2 \geq 4 \\ & y_1, y_2 \geq 0 \end{array}$$

(b). The optimal solution is (2, 1) and the optimal value is 18.

(c). We first notice that the second constraint in the dual problem is not tight: $3y_1^* + 2y_2^* - 1 \neq 0$, which implies $x_2^* = 0$.
Then, we observe that the first and the third constraints are tight in the dual problem: $y_1^* + y_3^* - 3 = 0, y_1^* + 2y_2^* - 4 = 0$, therefore $y_1^* \neq 0, y_2^* \neq 0$, we have that

$$\begin{aligned} x_1^* + x_3^* &= 5 \\ x_1^* + 2x_3^* &= 8 \end{aligned}$$

As a result, we get that $x_1^* = 2, x_3^* = 3$. Thus, the optimal solution of the primal problem is $(2, 0, 3)$.